



[GCT555 3DID]

ProxiWall: Seamless Private-to-Shared Mode Switching for Collaborative 3D Exploration on Dual-Wall Displays

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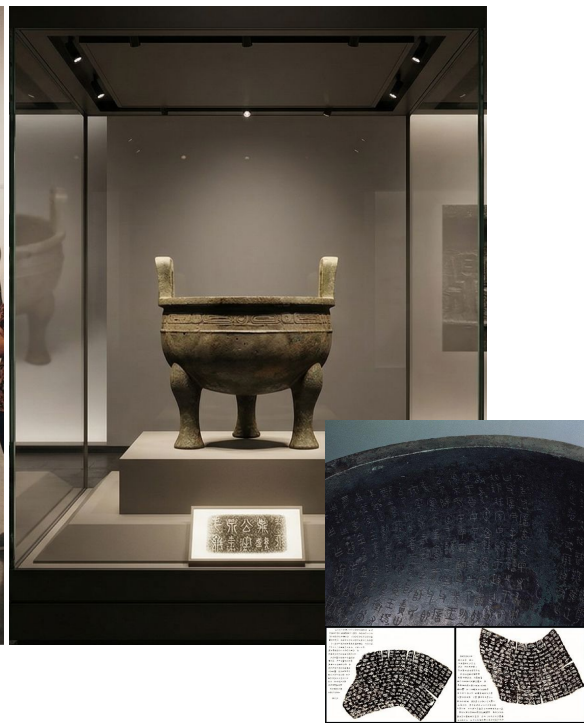


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Background





Needs (Target User)



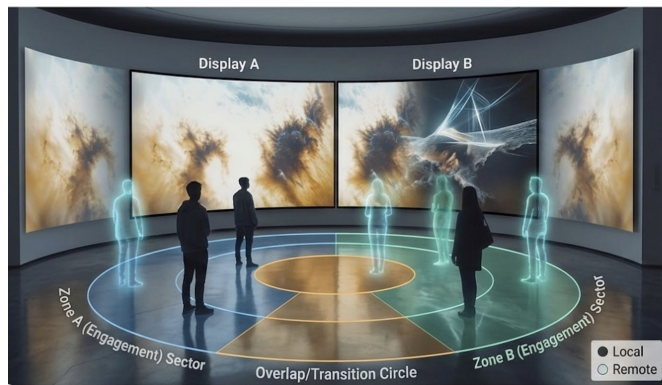


Needs (Target User)

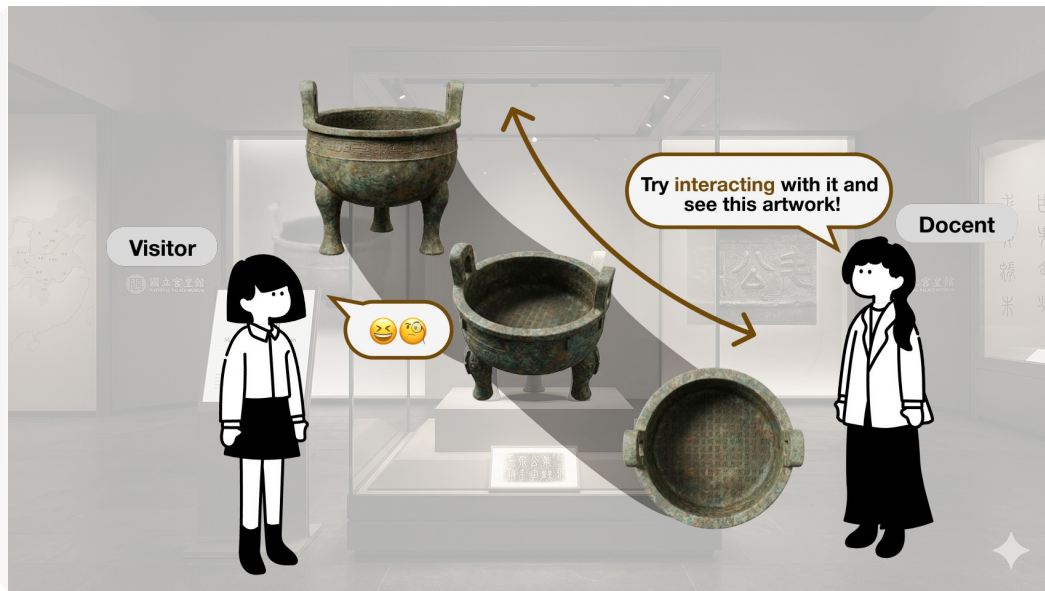




Approach



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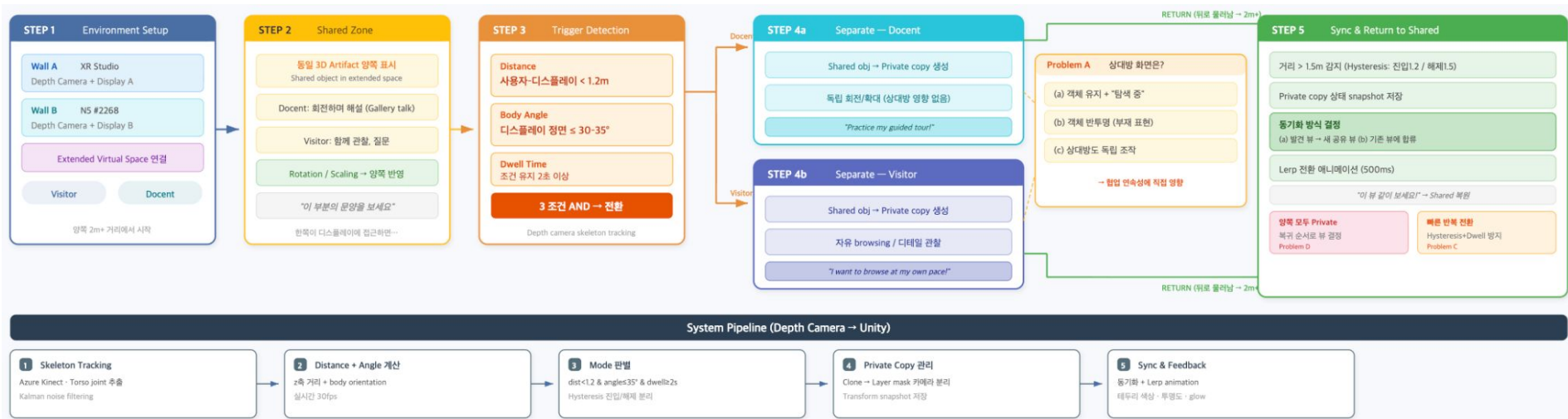


wall displays with spatial zones

each user's viewpoint angle to create a shared space



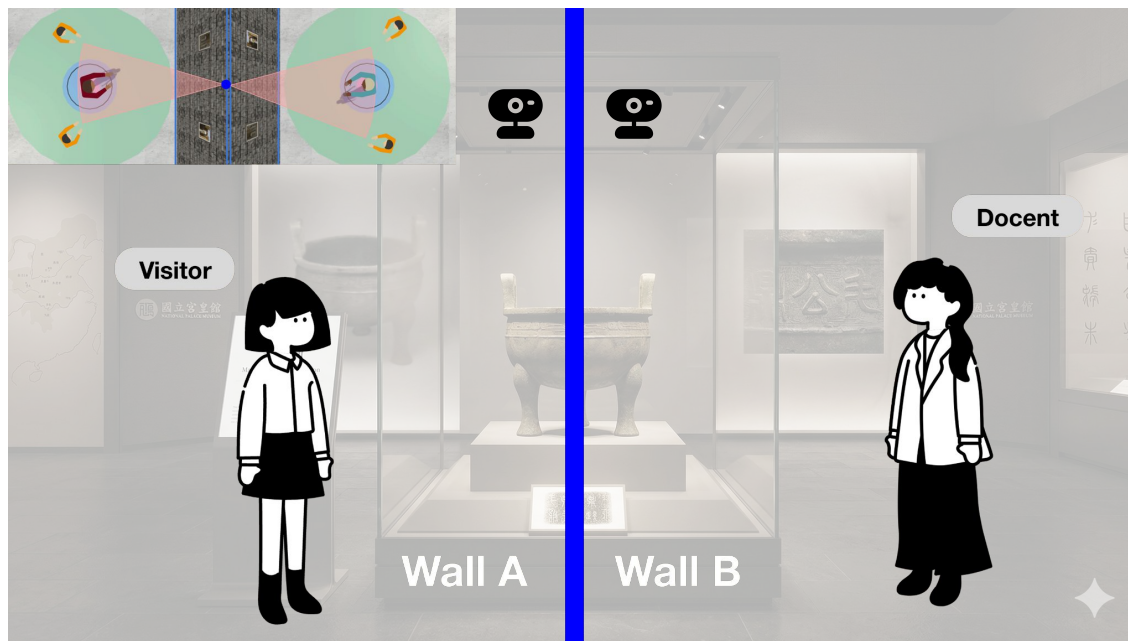
UX Scenario (Flow Chart)





UX Scenario

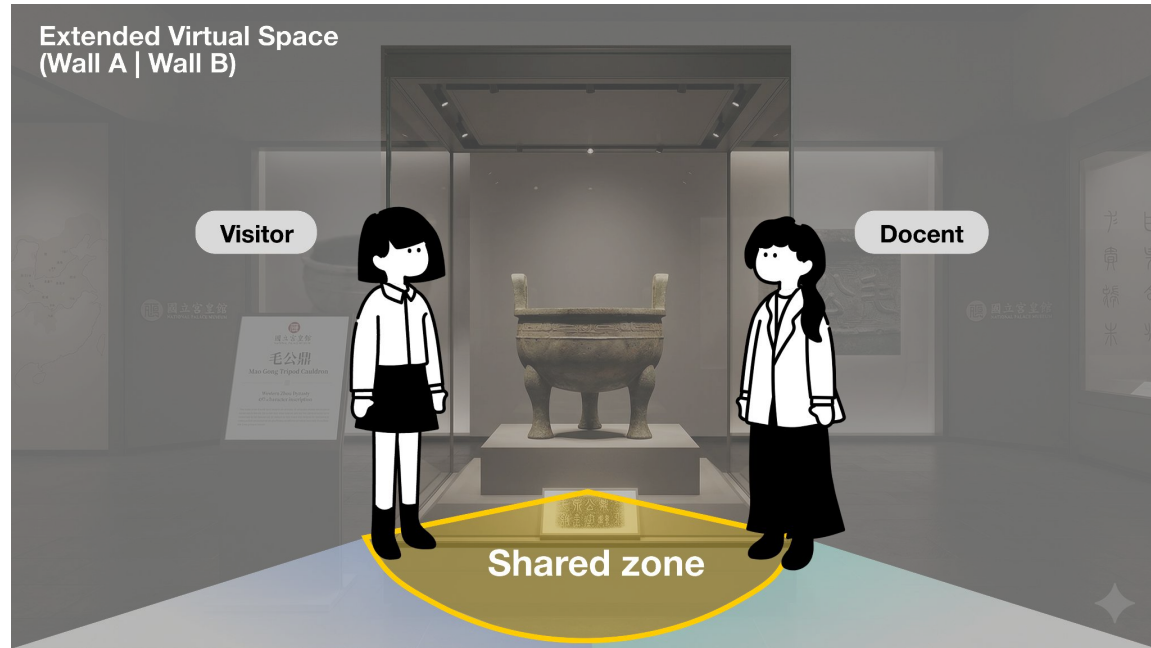
(1) Environment Setup : 2 Camera, Wall Display A (XR Studio), B (N5 #2268)





UX Scenario

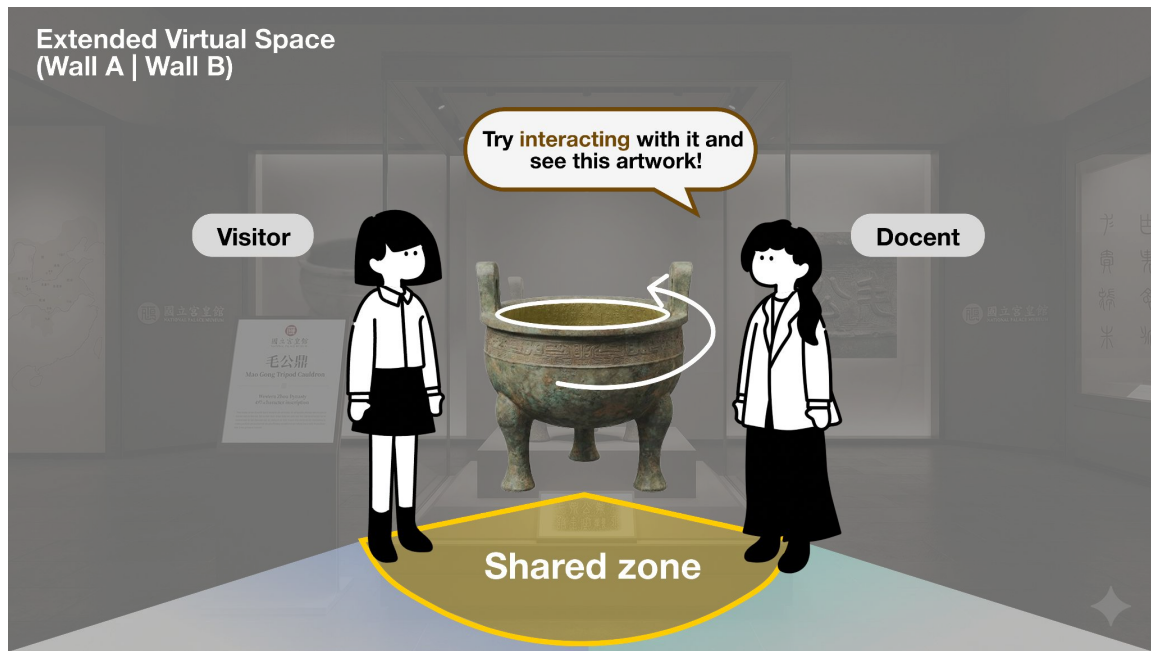
(2) Extended Virtual Space (Wall A | Wall B) -> Shared Zone : Walk into intersection zone.





UX Scenario

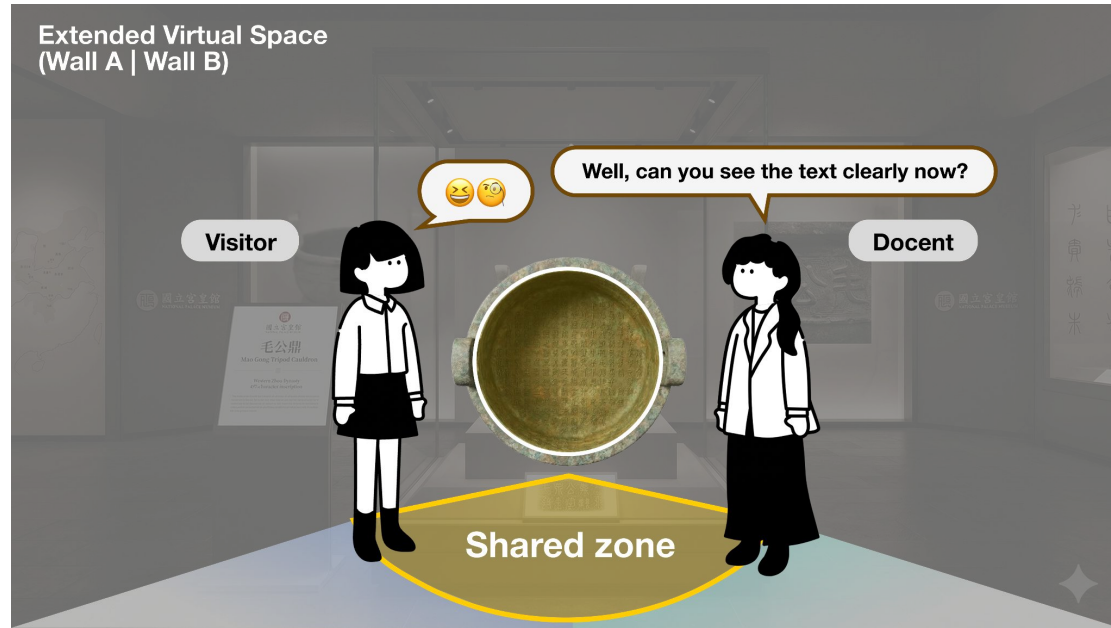
(3) Shared Zone : Seamlessly transition to 3D artwork exploration interaction without any explicit input





UX Scenario

(3) Shared Zone : 3D Artwork Exploration Interaction (Object Rotation, Scaling)



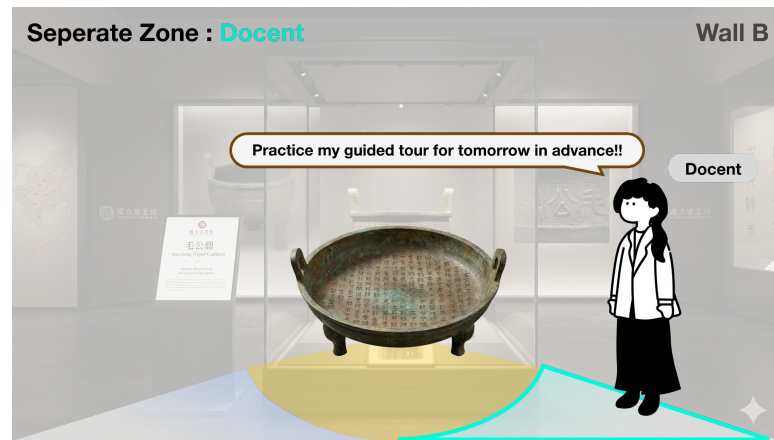


UX Scenario

(4) Separate Zone

Visitor : When moving beyond the viewpoint angle, the visitor (user) enters a separate zone.

Docent: Docent with collaborative interaction for guided tour.

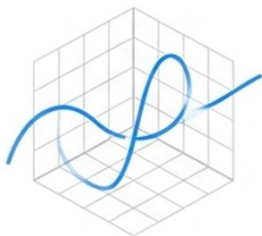




Benefit

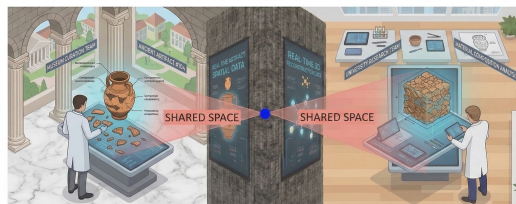
Expected results and 3 key contributions

Fluid Transition



Using natural body movement, we can examine the objects privately during an ongoing docent tour.

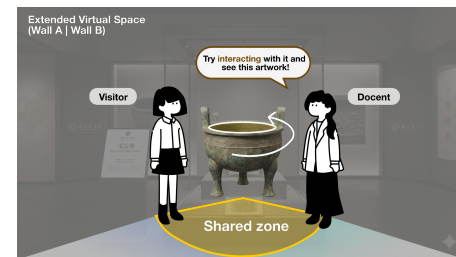
Dual-Wall Collaboration



Wall Display (A) Wall Display (B)

A wall with a depth camera enables smooth collaboration between docent and visitor.

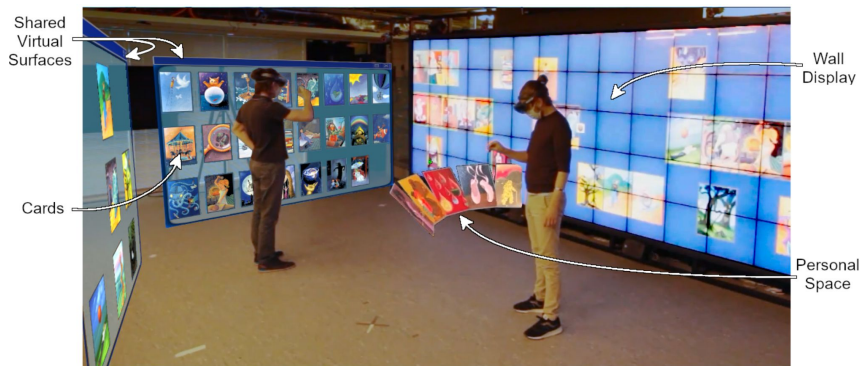
Empirical Design Guidelines



Empirical evidence from a user study on distance threshold, visual feedback strategies, and the impact of private exploration



(1) Comparison : Wall Displays + AR



Evaluating the Extension of Wall Displays with AR for Collaborative Work.

James, R., Bezerianos, A., & Chapuis, O. (2023, CHI).

Proposed Workspace System: Wall Displays + AR

Pros

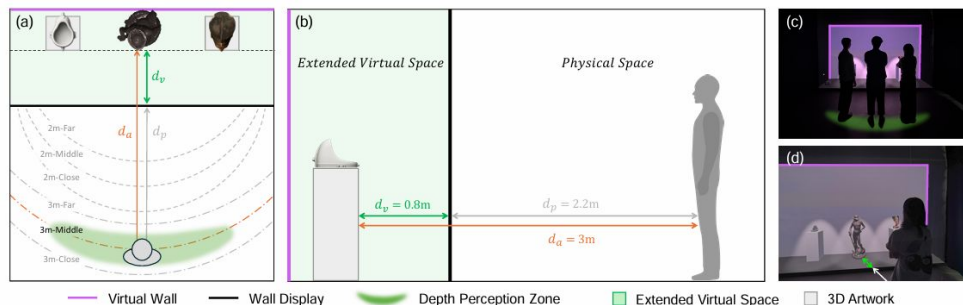
- Evaluated the combination of Wall Displays + AR against standalone Wall Display usage
- Wall Displays serve as Co-located Space, while AR functions as Personal Space
- Most participants preferred the Wall+AR condition — rated as more enjoyable and immersive

Cons

- Limited to collaborative scenarios where Wall Displays are used as a shared workspace
- No significant improvement in task performance
- Extends the workspace conceptually, not the user's physical space



(2) Comparison : Intersection zone using angle view point



Viewpoint-Tolerant Depth Perception for Shared Extended Space Experience on Wall-Sized Display

in IEEE Transactions on Visualization and Computer Graphics (ISMAR-TVCG / 2025)

Pros

- **Autonomy for Private Exploration:**
By simply stepping into the private zone, a user can independently manipulate the 3D object.
- **Systematic Management of Workspace Awareness:**
Controls the visual feedback during private mode to maintain a seamless flow of collaboration.

Cons

- **Potential Disruption of Co-presence:**
The private mode may negatively impact their sense of togetherness and engagement.



(3) Comparison : Shared space telepresence system



Project Starline: A high-fidelity telepresence system

in ACM SIGGRAPH 2024 Emerging Technologies, pp.1–2, 2024.1

Pros

Larger Space: The system provides a larger area where users can freely walk around and interact intuitively based on their physical distance.

Divided Private/Shared Space: The zones are clearly separated by the user's distance from the display. This allows users to control what they want to share with their partner and what they want to view independently.

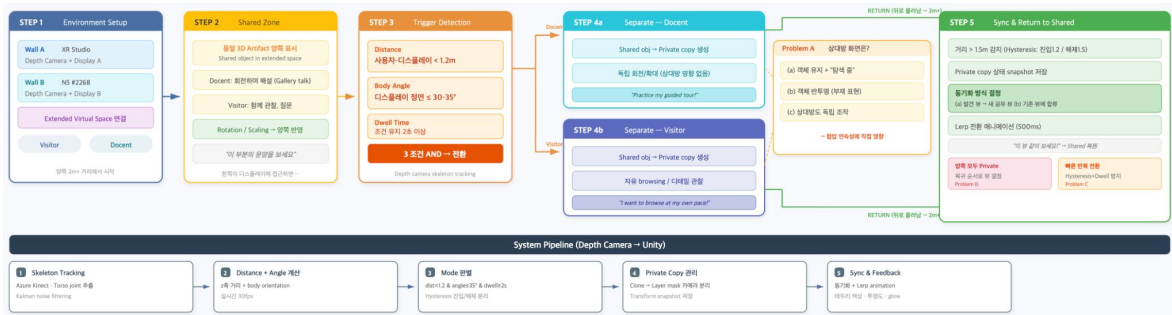
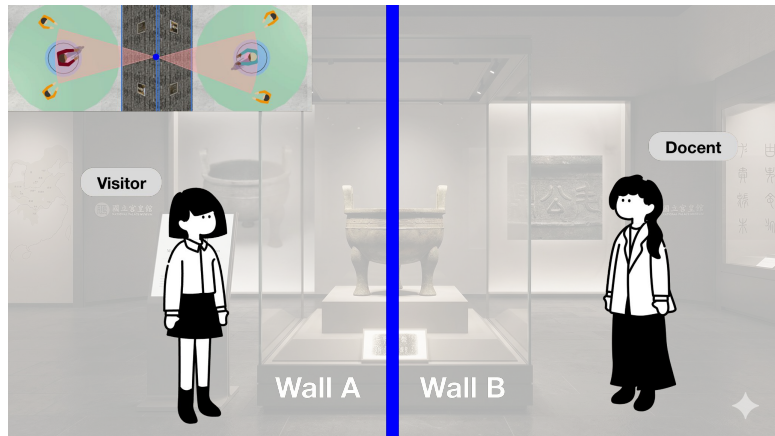
Cons

Immersion: The overall immersion and feeling of co-presence might be lower compared to systems like Starline, which provide a highly realistic illusion of a real person.



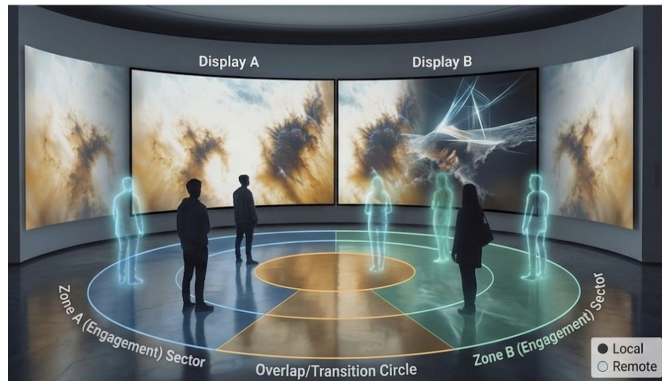
Deliverables

An interactive system for dual-wall display environments that leverages the user's physical distance as an implicit input to naturally transition between private and shared exploration modes.





Q&A



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